



Privacy-Preserving Speaker Verification via End-to-End Secure Representation Learning

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Abstract

The widespread adoption of cloud-based speaker verification raises privacy concerns, as unauthorized access to speaker voice can be exploited for fraudulent purposes. In response, privacy-preserving speaker verification (PPSV) schemes have emerged to safeguard speaker information and prevent unauthorized access. However, existing PPSV methods often compromise accuracy in their efforts to protect privacy. To enhance both privacy and accuracy, we propose an end-to-end framework for speaker template encryption that simultaneously optimizes verification and encryption tasks. Our approach enhances the security and accuracy of speaker verification by introducing quintuple loss and sharpness-aware minimization, respectively, eliminating texture details that could be exploited by attackers. We distill our methodologies into a PPSV method, Secure-RawNet. Experimental results demonstrate that Secure-RawNet achieves high accuracy and robust privacy protection.

Index Terms: privacy preservation, speaker verification, security, end-to-end

1. Introduction

As speaker verification technology becomes increasingly widespread, safeguarding the privacy of sensitive speaker embeddings has become a critical concern. During speaker verification, clients share their voices with a server, thereby exposing not only their conversations but also personal information that can be inferred from their voices. Since speaker embeddings are unique and private, using unencrypted speaker embeddings for authentication poses significant security risks. Moreover, storing speaker embeddings in the cloud further heightens privacy risks, as they can be misused to train voice cloning systems that mimic users' voices, posing a serious threat to personal privacy.

In response to growing security demands, the need for privacy-preserving speaker verification (PPSV) has become increasingly critical [1, 2, 3]. This technology aims to prevent the leakage of users' embeddings during the cloud-based speaker verification, ensuring that sensitive data remains secure. Recently, various PPSV schemes have been constructed to address the forementioned challenges, including random matrix [4], hash mapping [5] and homomorphic encryption [6].

However, these traditional PPSV methods still face challenges, primarily in effectively protecting speaker features. Even subtle residual features can be exploited by attackers, compromising security. Additionally, these methods typically require an acoustic feature extraction module prior to speaker embedding extraction. The reliance on artificially designed acoustic features, however, leads to information loss when com-

pared to directly using raw waveform input. Moreover, traditional approaches necessitate the design of a separate template encryption module based on hash functions or matrices. The use of these artificially defined encryption mechanisms further limits the preservation of speaker information. The additional acoustic feature extraction and encryption modules often negatively impacts verification accuracy, underscoring the need for exploring end-to-end (E2E) PPSV approaches.

To fill the gap, we propose a theoretically grounded and easily applicable framework for end-to-end PPSV, named Secure-RawNet. To achieve a balance between privacy and optimal recognizability, we reconfigure the acoustic and encryption modules while preserving their core functionalities. Additionally, we introduce a novel loss function and optimization strategy to enhance both speaker recognizability and privacy. Experimental results demonstrate that Secure-RawNet not only achieves high speaker verification accuracy but also offers improved privacy protection. Specifically, our main findings in this work cover three distinct domains:

- Conceptually, we propose a novel end-to-end PPSV framework that directly converts raw audio into encrypted embeddings which prevents the intermediate process from attacks. By introducing random information (key) unrelated to identity into the network, we address the challenge of encrypting speaker embeddings using neural networks.
- Technically, we develop a secure architecture named Secure-RawNet that realizes our conceptual idea. RawNet takes raw waveform as input, thereby avoiding the loss of phase or other information during feature extraction. We also introduce quintuplet loss, which enables cancelability of encrypted templates with easily changeable keys.
- Experimentally, we conduct a thorough evaluation of state-of-the-art PPSV works and demonstrate the feasibility of achieving both high privacy and satisfactory speaker verification performance. By extending the work of RawNet3 [7], we illustrate that sharpness-aware minimization can further push the limits of raw waveform speaker verification, particularly in terms of open-set generalizability.

The rest of the paper is organized as follows. Section 2 presents the proposed end-to-end PPSV method in detail. The experimental setup and results are presented in Section 3 and Section 4, respectively. Finally, Section 5 concludes this work.

2. Method

Current speaker verification systems identify individuals by comparing N-dimensional feature embeddings, known as speaker embeddings (templates). The widespread use of speaker verification has raised privacy concerns, as unautho-

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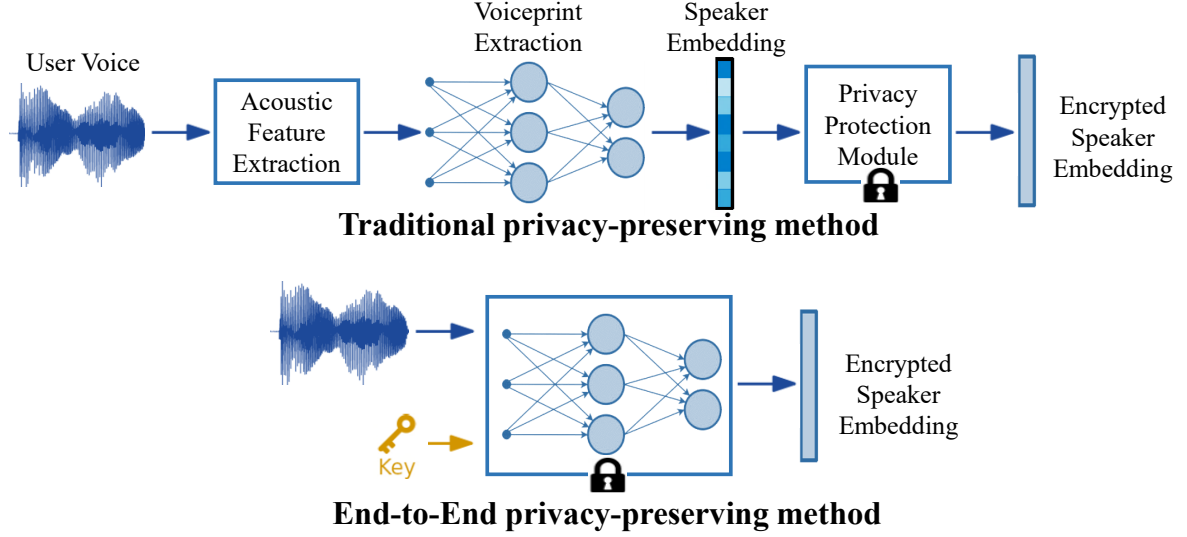


Figure 1: Diagram comparing traditional and end-to-end privacy-preserving speaker verification systems. The traditional method employs a cascading structure, which leads to multiple instances of information loss. Moreover, the use of a shared matrix key increases vulnerability to attacks. In contrast, the end-to-end privacy-preserving method enhances data security and user privacy while minimizing the impact on speaker verification accuracy.

rized access to speaker voice data can expose sensitive personal information. PPSV addresses these concerns by encoding the raw waveform x using a model θ , ensuring that private information cannot be inferred from the resulting speaker embedding $v = \theta(x)$, while still retaining sufficient utility for downstream speech tasks. In cases where a user's embedding is stolen, an unencrypted speaker verification model would require retraining, whereas a PPSV system would only necessitate a key change.

2.1. End-to-End PPSV

This paper introduces an end-to-end PPSV system to address the limitations of traditional methods. As depicted in Fig. 1, conventional PPSV approaches involve multiple stages, including acoustic feature extraction, speaker embedding extraction, and template encryption, each requiring manual design, fine-tuning, and domain-specific expertise. In contrast, the end-to-end model integrates these complex and labor-intensive stages into a single unified process, where all the feature extraction and encryption processes are automatically learned by the network. This allows server to calculate final output directly from the original input without manually defined intermediate steps.

The end-to-end PPSV method directly converts user recordings into encrypted speaker embeddings, reducing the risk of attacks on intermediate processes (acoustic features, unprotected embeddings), and lowering computing time and resource consumption. Besides, speaker independent random information is added to the model in the form of key vector k , ensuring that even if an intruder obtains the encrypted speaker embedding, the user's vocal characteristics cannot be reconstructed, maintaining the irreversibility of encryption process. Moreover, when the speaker embedding ciphertext is stolen, the system can be restored by simply replacing the key, eliminating the need to retrain model and ensuring the cancelability of the encryption process. We implement this approach using the Secure-RawNet architecture and Quintuplet Loss function, and further enhance accuracy with sharpness-aware minimization(SAM) [8, 9, 10].

2.2. Secure-RawNet Architecture

First, the RawNet3-based [11] speaker classification model is trained using AAM-Softmax loss [12] to ensure accurate speaker verification. The feature extraction layers of the trained RawNet3 model are then used as the feature extractor to obtain speaker embeddings from the input waveform.

Next, a key is introduced for security training, utilizing the Secure-RawNet network and quintuplet loss depicted in Fig. 2 to ensure the privacy protection of the feature template. In this stage, the speaker embeddings are concatenated with the key and passed through a fully connected layer. The network is optimized using quintuplet loss, which minimizes the distance between the pairs which match speaker and key while maximizing the distance between mismatched pairs.

2.3. Quintuplet Loss

The quintuplet loss is a novel variant of the widely-used triplet loss [13] for discriminative training. This approach enables the training of end-to-end neural networks for secure speaker verification. During training, in addition to the classic triplet inputs X_A , X_P and X_N , the network also receives two different keys (k_1 and k_2) bound to the input. The anchor X_A is only matched with correct key (k_1), $V_A = \theta(X_A, k_1)$. Similarly, the positive and negative samples are processed twice by the network, once with k_1 and once with k_2 , yielding four vectors:

$$\begin{aligned} V_{P1} &= \theta(X_P, k_1), V_{N1} = \theta(X_N, k_1), \\ V_{P2} &= \theta(X_P, k_2), V_{N2} = \theta(X_N, k_2). \end{aligned} \quad (1)$$

Then the distances between the anchor point V_A and four sample features V_{P1} , V_{P2} , V_{N1} , V_{N2} are calculated:

$$\begin{aligned} D_{P1} &= d(V_A, V_{P1}), D_{N1} = d(V_A, V_{N1}), \\ D_{P2} &= d(V_A, V_{P2}), D_{N2} = d(V_A, V_{N2}). \end{aligned} \quad (2)$$

As shown in Fig 2, Secure-RawNet receives five inputs (quintuplets) at once and optimizes the network by minimizing the

quintuplet loss function (minimizing the distance D_{P1} from the positive samples to anchor and maximizing the distance D_{N1} , D_{N2} and D_{P2} from negative samples to anchor). Additionally, a margin constraint α is applied to ensure a minimum margin α between the positive and negative pair distances. Two variants of quintuplet loss are defined and compared in our experiments:

$$\mathcal{L}_1^{quin}(\theta) = \max(D_{P1} - \min(D_{P2}, D_{N1}, D_{N2}) + \alpha, 0). \quad (3)$$

$$\begin{aligned} \mathcal{L}_2^{quin}(\theta) = & \max(D_{P1} - D_{P2} + \alpha, 0) \\ & + \max(D_{P1} - D_{N1} + \alpha, 0) \\ & + \max(D_{P1} - D_{N2} + \alpha, 0). \end{aligned} \quad (4)$$

These losses make sure that the representation that matches the identity and key is closer to the anchor in the projection space than the other three representations. Quintuplet loss enables using key vector to protect the speaker template without requiring any additional steps outside of neural network. As a result, this method fully leverages the capabilities of the end-to-end network while ensuring the security of speaker embeddings.

2.4. Sharpness-Aware Minimization

To achieve better open-set speaker verification performance, it is often preferable for an optimizer to converge to flat minima, which are characterized by uniformly low loss values, rather than sharp minima with abrupt loss changes. We adopt sharpness-aware minimization (SAM) for this purpose, with the objective function formulated as follows:

$$\min_{\theta} \max_{\|\epsilon\|_2 \leq \rho} \mathcal{L}(\theta + \epsilon) + \beta \|\theta\|_2^2, \quad (5)$$

where ρ denotes the radius of the L_2 norm ball centered at the model parameters θ . By minimizing both training loss and loss sharpness simultaneously, SAM-based optimizer seeks models situated in neighborhoods characterized by consistently low loss values, resulting in better generalization for open-set speaker verification.

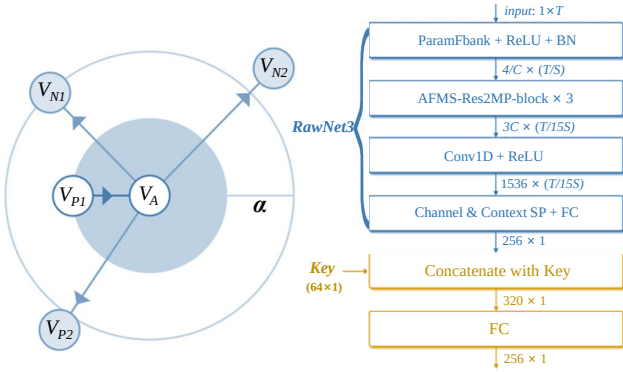


Figure 2: Left: Quintuplet Loss, which promotes the proximity between positive sample V_{P1} and anchor representation V_A , while pushing three negative samples (V_{P2} , V_{N1} , V_{N2}) away. Right: Secure-RawNet building block. During the first stage of training, only the RawNet3 network (blue) is optimized. In the second stage, the parameters of RawNet3 are frozen, and the key-related layers (yellow) are optimized using quintuplet loss.

3. Experiments

During training, the key is represented as a random vector, enabling users to specify a custom key during enrollment after

model training. If no key is provided during users' enrollment, the system assigns a random key by default. This randomness ensures that the model learns the relationship between the key and the speaker rather than relying on a fixed key pattern. After training Secure-RawNet, users can set an arbitrary key during registration and modify it freely in response to potential attacks, without requiring model retraining.

3.1. Dataset and Configuration

We utilized both the development portions of VoxCeleb1 and VoxCeleb2 [14, 15], whereas, for the evaluation, the Vox1-O test trials were employed. To enhance model robustness, we applied online data augmentation techniques [16] with MUSAN [17] and RIR datasets [18]. Additionally, speed perturbation [19] with 0.9 and 1.1 times was used to make the number of speakers in the training dataset tripled.

For each utterance, a 300-frame segment was randomly selected. No pre-processing, such as acoustic feature extraction or normalization, was applied to the raw waveforms. Training was conducted using PyTorch, employing SAM-based Adam/AdamW optimizer [20, 21, 22]. The learning rate, scheduled between $5e-4$ and $3e-6$, followed a cosine annealing strategy [23] over 60 epochs. We use cosine distance as the similarity criterion with scores calibrated using adaptive score normalization (AS-Norm) with an imposter cohort size of 600.

3.2. Evaluation Metrics

The experiments are designed to evaluate the performance of the models trained with quintuplet loss, not only considering verification accuracy, but also speaker security. Measuring how much information an observation leaks about a secret quantity it depends on is a classical question that has long been studied in information theory. We select R_I [24] as the main metric of security, which can be expressed as:

$$R_I = \frac{I(X; Y)}{H(X)}, \quad (6)$$

where $I(X; Y)$ is the mutual information of unprotected template X and the protected template Y , and $H(X)$ is the entropy of X . Low R_I means even when one has all knowledge of v , obtaining information on x should be impossible. Speaker verification accuracy is reported using the equal error rate (EER) and minimum detection cost function (min-DCF) [25].

4. Results and Discussion

A general overview of the results obtained is presented in Table 1 for accuracy and security. The following subsections discuss the results on radius ρ of SAM-based optimizer, margin, dimension of key vector and number of quintuplets.

4.1. Speaker Verification Performance

The results of speaker verification are presented in Table 1, which demonstrate that the proposed quintuplet loss effectively optimizes end-to-end privacy-preserving speaker verification models. Both $\mathcal{L}_1^{quin}$ and $\mathcal{L}_2^{quin}$ yielded relatively low EERs indicating the effectiveness of the end-to-end approach. Secure-RawNet with loss $\mathcal{L}_2^{quin}$ achieved an EER of 0.7872%, marking a significant performance improvement over the traditional PPSV methods. In contrast, previous methods (secure

multiparty and attribute- or matrix-based¹ approaches), exhibit higher EERs and greater information leakage, underscoring the privacy risks associated with these techniques. Our proposed system successfully encodes raw waveform into secure speaker templates without revealing privacy-sensitive information.

Further analysis of verification scores, as shown in the histogram in Figure 3 (a), reveals distinct score distributions for target and non-target trials. Target scores exhibit a peak distribution, while non-target scores display a smoother distribution due to mismatches in key or identity. Without forcing the model to adhere to predefined encodings during training, the learned representations exhibit intra-class compactness and diversity between the same user’s enrollment and various speaker verification attempts.

Table 1: *Speaker verification and secrecy leakage evaluation*

Method	EER(%)↓	min-DCF↓	R_I (%)↓
Gaussian Matrix [26]	1.1259	0.077	1.04
Random Matrix [4]	0.9154	0.066	1.52
Secure Multiparty [27]	3.2	0.205	-
Attribute Driven [28]	2.36	0.154	-
Secure-RawNet ($\mathcal{L}_1^{quin}$)	1.0228	0.074	0.39
Secure-RawNet ($\mathcal{L}_2^{quin}$)	0.7872	0.065	0.31

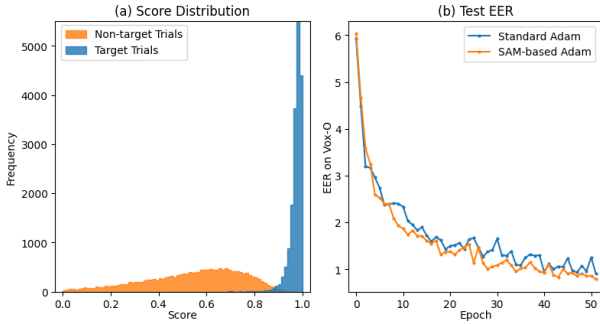


Figure 3: (a) *Score distribution of Secure-RawNet on Vox1-O set.* (b) *The EER evolution with standard Adam and SAM-based Adam on Vox1-O set.*

4.2. Privacy Preservation Performance

Regarding the privacy preservation and secrecy leakage evaluation, the safety metrics are summarized in Table 1. Similarly, both $\mathcal{L}_1^{quin}$ and $\mathcal{L}_2^{quin}$ losses resulted in low secrecy leakage (R_I), indicating that the keys used to make the templates cancelable reveal no information on them. This suggests that it is highly challenging for an attacker to recover the original speaker embedding x from the compromised templates v generated by the Secure-RawNet model. The optimized privacy-preserving model compresses the inputs retaining only the information needed for speaker related tasks.

These findings on mutual information and secrecy leakage highlight the significant advantages of using E2E deep learning models for privacy preservation. Neural network is used not only to model the speaker embedding space, but also privacy protection algorithm and acoustic feature extraction mod-

¹We found no open-source code for two matrix-based methods. We reproduce them to our best effort, recognizing the possibility of inconsistencies.

ules. By integrating information security with similarity measurement techniques, our proposed PPSV system effectively ensures the security of speaker templates.

4.3. Ablation study

An ablation study was conducted on the proposed system’s parameters to assess their impact. As shown in Table 2, incorporating SAM resulted in a relative EER reduction of 11.37% and 13.19% for Adam and AdamW, respectively, indicating significantly improved generalization. Additionally, SAM also optimized convergence speed and enhanced training stability, as illustrated in Figure 3 (b). While performance may vary depending on specific scenarios, a ρ value between 0.01 and 0.03 is expected to yield the highest likelihood of success.

Table 2: *Evaluation on speaker verification with SAM optimizer*

Model	With SAM	Param ρ	EER(%)↓	
			Adam	AdamW
RawNet3 [7]	-	-	0.8882	1.1701
Secure-RawNet	✓	0.005	0.8084	1.1514
Secure-RawNet	✓	0.010	0.7905	1.0903
Secure-RawNet	✓	0.030	0.8084	1.0158
Secure-RawNet	✓	0.050	0.8616	1.0793

A major caveat of the quintuplet loss, though, is that as the dataset gets larger, the number of quintuplets (N_q) grows cubically, rendering a long enough training impractical. Table 3 presents the EER and R_I obtained by varying parameters that balance the security and accuracy. Notably, varying the margin (α) and key dimension (D_k) affects performance, with $\alpha = 0.3$ and $D_k = 64$ yielding the lowest EER values. The results underscore the significance of margin and key dimension, with the suitable configuration leading to superior safety and precision.

Table 3: *Effects of margin and the dimension of key*

Margin	Dimension _{key}	EER(%)↓	min-DCF↓
0	64	1.5734	0.109
0.3	64	0.7905	0.065
0.6	64	0.8561	0.067
1	64	0.8883	0.067
0.3	32	0.8229	0.063
0.3	64	0.7905	0.065
0.3	128	0.8602	0.064

5. Conclusion

The end-to-end privacy-preserving speaker verification system proposed in this paper achieves both enhanced security and accuracy through a learnable encryption. The neural network is utilized not only to model the speaker embedding space but also to incorporate privacy protection algorithms and acoustic feature extraction. By using an end-to-end network to learn the privacy protection algorithm, rather than explicitly defining it, the model is able to develop a more secure speaker embedding space. Experimental results and analysis demonstrate the effectiveness and generalizability of the proposed Secure-RawNet model, highlighting its superiority over traditional privacy-preserving methods in the literature. Moreover, this work identifies accuracy vulnerabilities in evaluated state-of-the-art PPSV schemes, providing a foundation for the development of more effective countermeasures.

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